Assignment 1

Task 3: Database Implementation

Group Member Names:

Madhavi Latha Tripuram - C3528532

Hari Krishna Jadda - C3511138

Dhimanshi Piripun Jayasena - C3503338

Lab Timings: **Thursday 1.00pm to 3.00pm**

Course coordinator: **Prof. Mahakim Newton**

Database Creation Script

```
/*
GlobalCarHire Database Creation Script
Authors: Dhimanshi Piripun Jayasena, Hari Krishna Jadda, Madhavi Latha Tripuram
*/

-- Create the database
CREATE DATABASE GlobalCarHire;
GO

USE GlobalCarHire;
GO

-- Create Branch table
CREATE TABLE Branch (
    branchId VARCHAR(10) NOT NULL,
    name VARCHAR(50) NOT NULL,
```

```
    address VARCHAR(100) NOT NULL,
    country VARCHAR(50) NOT NULL,
    phoneNumber VARCHAR(15) NOT NULL,
    currency VARCHAR(3) NOT NULL,
    operationalStatus VARCHAR(10) NOT NULL
        CHECK (operationalStatus IN ('Active', 'Inactive', 'Closed'))
        DEFAULT 'Active',

    CONSTRAINT PK_Branch PRIMARY KEY (branchId),
    CONSTRAINT UQ_Branch_Phone UNIQUE (phoneNumber)
);
GO

-- Create VehicleCategory table
CREATE TABLE VehicleCategory (
    categoryId VARCHAR(10) NOT NULL,
    name VARCHAR(20) NOT NULL,
    description VARCHAR(100),
    dailyRateMultiplier DECIMAL(3,2) NOT NULL DEFAULT 1.00,

    CONSTRAINT PK_VehicleCategory PRIMARY KEY (categoryId),
    CONSTRAINT UQ_VehicleCategory_Name UNIQUE (name),
    CONSTRAINT CHK_DailyRateMultiplier CHECK (dailyRateMultiplier > 0)
);
GO

-- Create Employee superclass table
CREATE TABLE Employee (
    employeeId VARCHAR(10) NOT NULL,
    name VARCHAR(50) NOT NULL,
```

```
    position VARCHAR(10) NOT NULL
        CHECK (position IN ('Manager', 'Associate')),
    branchId VARCHAR(10) NOT NULL,

    CONSTRAINT PK_Employee PRIMARY KEY (employeeId),
    CONSTRAINT FK_Employee_Branch FOREIGN KEY (branchId)
        REFERENCES Branch(branchId)
        ON UPDATE CASCADE
        ON DELETE CASCADE
);
GO

-- Create ManagerDetails subclass table
CREATE TABLE ManagerDetails (
    employeeId VARCHAR(10) NOT NULL,
    managerLevel VARCHAR(20) NOT NULL,
    approvalAuthority VARCHAR(50),

    CONSTRAINT PK_ManagerDetails PRIMARY KEY (employeeId),
    CONSTRAINT FK_ManagerDetails_Employee FOREIGN KEY (employeeId)
        REFERENCES Employee(employeeId)
        ON UPDATE CASCADE
        ON DELETE CASCADE
);
GO

-- Create AssociateDetails subclass table
CREATE TABLE AssociateDetails (
    employeeId VARCHAR(10) NOT NULL,
    specialization VARCHAR(50),
```

```
    CONSTRAINT PK_AssociateDetails PRIMARY KEY (employeeId),
    CONSTRAINT FK_AssociateDetails_Employee FOREIGN KEY (employeeId)
        REFERENCES Employee(employeeId)
        ON UPDATE CASCADE
        ON DELETE CASCADE
);
GO

-- Create Customer table
CREATE TABLE Customer (
    customerId VARCHAR(10) NOT NULL,
    fullName VARCHAR(50) NOT NULL,
    contactDetails VARCHAR(50) NOT NULL,
    email VARCHAR(50) NOT NULL,
    licenseInformation VARCHAR(50) NOT NULL,
    loyaltyTier VARCHAR(10)
        CHECK (loyaltyTier IN ('Silver', 'Gold', 'Platinum'))
        DEFAULT 'Silver',

    CONSTRAINT PK_Customer PRIMARY KEY (customerId),
    CONSTRAINT UQ_Customer_Email UNIQUE (email),
    CONSTRAINT UQ_Customer_License UNIQUE (licenseInformation)
);
GO

-- Create Service superclass table
CREATE TABLE Service (
    serviceId VARCHAR(10) NOT NULL,
    name VARCHAR(50) NOT NULL,
```

```
    description VARCHAR(100),
    status VARCHAR(10) NOT NULL
        CHECK (status IN ('Available', 'Unavailable'))
        DEFAULT 'Available',
    serviceType VARCHAR(20) NOT NULL
        CHECK (serviceType IN ('Regular', 'Package')),

    CONSTRAINT PK_Service PRIMARY KEY (serviceId)
);
GO

-- Create RegularServiceDetails subclass table
CREATE TABLE RegularServiceDetails (
    serviceId VARCHAR(10) NOT NULL,
    basePrice DECIMAL(10,2) NOT NULL,
    currency VARCHAR(3) NOT NULL,
    restrictions VARCHAR(100),

    CONSTRAINT PK_RegularServiceDetails PRIMARY KEY (serviceId),
    CONSTRAINT FK_RegularServiceDetails_Service FOREIGN KEY (serviceId)
        REFERENCES Service(serviceId)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT CHK_RegularBasePrice CHECK (basePrice >= 0)
);
GO

-- Create PackageServiceDetails subclass table
CREATE TABLE PackageServiceDetails (
    serviceId VARCHAR(10) NOT NULL,
```

```
    basePrice DECIMAL(10,2) NOT NULL,
    currency VARCHAR(3) NOT NULL,
    validPeriodStart DATE NOT NULL,
    validPeriodEnd DATE NOT NULL,
    inclusions TEXT,
    exclusions TEXT,
    gracePeriod INT NOT NULL DEFAULT 1,

    CONSTRAINT PK_PackageServiceDetails PRIMARY KEY (serviceId),
    CONSTRAINT FK_PackageServiceDetails_Service FOREIGN KEY (serviceId)
        REFERENCES Service(serviceId)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT CHK_PackageBasePrice CHECK (basePrice >= 0),
    CONSTRAINT CHK_PackageValidPeriod CHECK (validPeriodEnd >=
validPeriodStart),
    CONSTRAINT CHK_GracePeriod CHECK (gracePeriod >= 0)
);
GO

-- Create Vehicle table
CREATE TABLE Vehicle (
    vehicleId VARCHAR(10) NOT NULL,
    make VARCHAR(20) NOT NULL,
    model VARCHAR(20) NOT NULL,
    year INT NOT NULL,
    registrationNumber VARCHAR(15) NOT NULL,
    status VARCHAR(15) NOT NULL
        CHECK (status IN ('Active', 'Inactive', 'Decommissioned'))
        DEFAULT 'Active',
    mileage INT NOT NULL DEFAULT 0,
```

```
    seatingCapacity INT NOT NULL,
    fuelType VARCHAR(10) NOT NULL
        CHECK (fuelType IN ('Diesel', 'Petrol', 'Electric')),
    transmissionType VARCHAR(10) NOT NULL
        CHECK (transmissionType IN ('Manual', 'Automatic')),
    baseRentalRate DECIMAL(10,2) NOT NULL,
    branchId VARCHAR(10) NOT NULL,
    categoryId VARCHAR(10) NOT NULL,

    CONSTRAINT PK_Vehicle PRIMARY KEY (vehicleId),
    CONSTRAINT UQ_Vehicle_Registration UNIQUE (registrationNumber),
    CONSTRAINT FK_Vehicle_Branch FOREIGN KEY (branchId)
        REFERENCES Branch(branchId)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT FK_Vehicle_Category FOREIGN KEY (categoryId)
        REFERENCES VehicleCategory(categoryId)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT CHK_VehicleYear CHECK (year BETWEEN 2000 AND
YEAR(GETDATE())),
    CONSTRAINT CHK_Mileage CHECK (mileage >= 0),
    CONSTRAINT CHK_SeatingCapacity CHECK (seatingCapacity > 0),
    CONSTRAINT CHK_BaseRentalRate CHECK (baseRentalRate >= 0)
);
GO

-- Create PricedService table
CREATE TABLE PricedService (
    pricedServiceId VARCHAR(10) NOT NULL,
    price DECIMAL(10,2) NOT NULL,
```

```
    pricingStartDate DATE NOT NULL,
    pricingEndDate DATE NOT NULL,
    approverManagerId VARCHAR(10) NOT NULL,
    serviceId VARCHAR(10) NOT NULL,
    branchId VARCHAR(10) NOT NULL,

    CONSTRAINT PK_PricedService PRIMARY KEY (pricedServiceId),
    CONSTRAINT FK_PricedService_Service FOREIGN KEY (serviceId)
        REFERENCES Service(serviceId)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT FK_PricedService_Branch FOREIGN KEY (branchId)
        REFERENCES Branch(branchId)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT FK_PricedService_Manager FOREIGN KEY (approverManagerId)
        REFERENCES Employee(employeeId),
    CONSTRAINT CHK_PricingPeriod CHECK (pricingEndDate >= pricingStartDate),
    CONSTRAINT CHK_Price CHECK (price >= 0)
);

-- Create ExtraService table
CREATE TABLE ExtraService (
    extraServiceId VARCHAR(10) NOT NULL,
    name VARCHAR(50) NOT NULL,
    description VARCHAR(100),
    advertisedRate DECIMAL(10,2) NOT NULL,

    CONSTRAINT PK_ExtraService PRIMARY KEY (extraServiceId),
    CONSTRAINT CHK_AdvertisedRate CHECK (advertisedRate >= 0)
```

```
);
GO

-- Create Driver table
CREATE TABLE Driver (
    driverId VARCHAR(10) NOT NULL,
    name VARCHAR(50) NOT NULL,
    licenseInformation VARCHAR(50) NOT NULL,
    contactNumbers VARCHAR(50),

    CONSTRAINT PK_Driver PRIMARY KEY (driverId),
    CONSTRAINT UQ_Driver_License UNIQUE (licenseInformation)
);
GO

-- Create Booking table
CREATE TABLE Booking (
    bookingId VARCHAR(10) NOT NULL,
    modeOfBooking VARCHAR(10) NOT NULL
        CHECK (modeOfBooking IN ('Website', 'App', 'Phone', 'In-person')),
    bookingDate DATETIME NOT NULL DEFAULT GETDATE(),
    customerInformation VARCHAR(10) NOT NULL,
    serviceStartDate DATE NOT NULL,
    serviceEndDate DATE NOT NULL,
    paymentMethod VARCHAR(20) NOT NULL,
    bookingDeposit DECIMAL(10,2) DEFAULT 0,
    bookingStatus VARCHAR(10) NOT NULL
        CHECK (bookingStatus IN ('Confirmed', 'Cancelled', 'Serviced'))
        DEFAULT 'Confirmed',
    customerId VARCHAR(10) NOT NULL,
```

```
    CONSTRAINT PK_Booking PRIMARY KEY (bookingId),
    CONSTRAINT FK_Booking_Customer FOREIGN KEY (customerId)
        REFERENCES Customer(customerId)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT CHK_BookingPeriod CHECK (serviceEndDate >=
serviceStartDate),
    CONSTRAINT CHK_BookingDeposit CHECK (bookingDeposit >= 0)
);
GO

-- Create VehicleTransaction table
CREATE TABLE VehicleTransaction (
    transactionId VARCHAR(10) NOT NULL,
    transactionType VARCHAR(10) NOT NULL
        CHECK (transactionType IN ('Pickup', 'Return')),
    actualDateTime DATETIME NOT NULL,
    vehicleCondition TEXT NOT NULL,
    mileageReading INT NOT NULL,
    fuelLevel VARCHAR(10) NOT NULL
        CHECK (fuelLevel IN ('Empty', 'Low', 'Half', 'Full')),
    additionalCharges DECIMAL(10,2) DEFAULT 0,
    bookingId VARCHAR(10) NOT NULL,
    vehicleId VARCHAR(10) NOT NULL,
    employeeId VARCHAR(10) NOT NULL,

    CONSTRAINT PK_VehicleTransaction PRIMARY KEY (transactionId),
    CONSTRAINT FK_VehicleTransaction_Booking FOREIGN KEY (bookingId)
        REFERENCES Booking(bookingId)
        ON UPDATE CASCADE
```

```
    ON DELETE CASCADE,
  CONSTRAINT FK_VehicleTransaction_Vehicle FOREIGN KEY (vehicleId)
    REFERENCES Vehicle(vehicleId)
    ON UPDATE CASCADE
    ON DELETE CASCADE,
  CONSTRAINT FK_VehicleTransaction_Employee FOREIGN KEY (employeeId)
    REFERENCES Employee(employeeId),
  CONSTRAINT CHK_MileageReading CHECK (mileageReading >= 0),
  CONSTRAINT CHK_AdditionalCharges CHECK (additionalCharges >= 0)
);
GO

-- Create Invoice table
CREATE TABLE Invoice (
  invoiceId VARCHAR(10) NOT NULL,
  baseRentalCharges DECIMAL(10,2) NOT NULL,
  extraServiceCharges DECIMAL(10,2) DEFAULT 0,
  totalLatePenalty DECIMAL(10,2) DEFAULT 0,
  totalDamagePenalty DECIMAL(10,2) DEFAULT 0,
  totalDiscount DECIMAL(10,2) DEFAULT 0,
  finalAmount DECIMAL(10,2) NOT NULL,
  paymentMethod VARCHAR(20) NOT NULL,
  bookingId VARCHAR(10) NOT NULL,

  CONSTRAINT PK_Invoice PRIMARY KEY (invoiceId),
  CONSTRAINT FK_Invoice_Booking FOREIGN KEY (bookingId)
    REFERENCES Booking(bookingId)
    ON UPDATE CASCADE
    ON DELETE CASCADE,
  CONSTRAINT CHK_BaseRentalCharges CHECK (baseRentalCharges >= 0),
```

```
    CONSTRAINT CHK_ExtraServiceCharges CHECK (extraServiceCharges >= 0),
    CONSTRAINT CHK_LatePenalty CHECK (totalLatePenalty >= 0),
    CONSTRAINT CHK_DamagePenalty CHECK (totalDamagePenalty >= 0),
    CONSTRAINT CHK_Discount CHECK (totalDiscount >= 0),
    CONSTRAINT CHK_FinalAmount CHECK (finalAmount >= 0)
);
GO

-- Create junction tables for many-to-many relationships

-- Booking_PricedService table
CREATE TABLE Booking_PricedService (
    bookingId VARCHAR(10) NOT NULL,
    pricedServiceId VARCHAR(10) NOT NULL,

    CONSTRAINT PK_Booking_PricedService PRIMARY KEY (bookingId,
pricedServiceId),
    CONSTRAINT FK_Booking_PricedService_Booking FOREIGN KEY (bookingId)
        REFERENCES Booking(bookingId)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT FK_Booking_PricedService_PricedService FOREIGN KEY
(pricedServiceId)
        REFERENCES PricedService(pricedServiceId)
        ON UPDATE CASCADE
        ON DELETE CASCADE
);
GO
```

```
-- Booking_Vehicle table
CREATE TABLE Booking_Vehicle (
    bookingId VARCHAR(10) NOT NULL,
    vehicleId VARCHAR(10) NOT NULL,

    CONSTRAINT PK_Booking_Vehicle PRIMARY KEY (bookingId, vehicleId),
    CONSTRAINT FK_Booking_Vehicle_Booking FOREIGN KEY (bookingId)
        REFERENCES Booking(bookingId)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT FK_Booking_Vehicle_Vehicle FOREIGN KEY (vehicleId)
        REFERENCES Vehicle(vehicleId)
        ON UPDATE CASCADE
        ON DELETE CASCADE
);
GO

-- Booking_Driver table
CREATE TABLE Booking_Driver (
    bookingId VARCHAR(10) NOT NULL,
    driverId VARCHAR(10) NOT NULL,

    CONSTRAINT PK_Booking_Driver PRIMARY KEY (bookingId, driverId),
    CONSTRAINT FK_Booking_Driver_Booking FOREIGN KEY (bookingId)
        REFERENCES Booking(bookingId)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT FK_Booking_Driver_Driver FOREIGN KEY (driverId)
        REFERENCES Driver(driverId)
        ON UPDATE CASCADE
```

```
        ON DELETE CASCADE
);
GO

-- Booking_ExtraService table
CREATE TABLE Booking_ExtraService (
    bookingId VARCHAR(10) NOT NULL,
    extraServiceId VARCHAR(10) NOT NULL,
    quantity INT NOT NULL DEFAULT 1,

    CONSTRAINT PK_Booking_ExtraService PRIMARY KEY (bookingId,
extraServiceId),
    CONSTRAINT FK_Booking_ExtraService_Booking FOREIGN KEY (bookingId)
        REFERENCES Booking(bookingId)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT FK_Booking_ExtraService_ExtraService FOREIGN KEY
(extraServiceId)
        REFERENCES ExtraService(extraServiceId)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT CHK_Quantity CHECK (quantity > 0)
);
GO

PRINT 'GlobalCarHire database tables created successfully with all constraints.';
GO
```